EPISODE 4 COMPANION REPORT

Values Alignment

From principles to reliable systems: a practical and technical overview of how values are selected, encoded, learned, tested, governed, and revised.

Audience
Researchers, product leaders, evaluators, and governance teams

Research cutoff
October 2026

Alignment succeeds only when the whole chain holds

Values alignment is not a single training technique or a benchmark score. It is a lifecycle discipline that connects legitimate value selection to system behavior under pressure, uncertainty, and change.

CORE IDEA

Separate the layers

Distinguish values, rules, training signals, learned behavior, deployment controls, and observed outcomes. Conflating them makes failures hard to diagnose.

DESIGN PRINCIPLE

Keep traceability

Every important behavior test should point back to a principle, an interpretation, an owner, and a decision about acceptable trade-offs.

TECHNICAL POSTURE

Use a portfolio

Instruction tuning, preference optimization, constitutional methods, interpretability, red teaming, and monitoring address different failure modes.

GOVERNANCE POSTURE

Make revision possible

Values change, contexts vary, and errors surface after deployment. Amendment, appeal, rollback, and incident learning are alignment mechanisms.

Bottom line. A well-aligned system is not one that always produces a preferred answer. It is one whose objectives, boundaries, uncertainty behavior, oversight, and revision processes remain defensible across contexts.

Six conclusions

1. **Values are inputs, not discoveries.** Data can reveal preferences; it cannot by itself determine which preferences deserve authority.
2. **Preference fit is not moral correctness.** Human feedback can reward fluency, agreement, or convenience over truth and fairness.
3. **Generalization is the real test.** Training-time behavior is weak evidence when deployment introduces new stakes, users, tools, and incentives.
4. **Benchmarks need counterfactuals.** Test where proxies diverge from the underlying value, not only where they correlate.
5. **Deployment architecture matters.** Permissions, escalation, logging, and rollback can reduce harm even when model alignment is imperfect.
6. **Legitimacy is measurable work.** Provenance, participation, conflict rules, and contestability should be documented and audited.

A strong score can hide a broken alignment chain

A model can satisfy a test while failing the value the test was meant to represent. Diagnose alignment by locating the broken link.

01	02	03	04	05	06
Select	**Specify**	**Teach**	**Learn**	**Verify**	**Govern**
Whose values count, under what authority, for which domain?	Translate abstract values into rules, priorities, and exceptions.	Create demonstrations, comparisons, critiques, and environments.	Optimize behavior without overfitting a proxy or reward model.	Test conflicts, shifts, adversaries, and uncertainty.	Monitor outcomes, handle appeals, revise, and roll back.

Three meanings of alignment

- **Intent alignment:** the system follows the user's legitimate request as understood.
- **Preference alignment:** outputs match stated or inferred preferences of a person or population.
- **Value alignment:** behavior respects justified norms, rights, duties, and social constraints, including when preferences conflict.

These overlap but are not equivalent. A system can follow an instruction that violates a policy, match a preference that is factually wrong, or obey a rule whose application is unjust.

Outer and inner questions

Outer alignment: Is the specified objective a good proxy for what designers and stakeholders actually want?

Inner alignment: Did training produce a system that robustly pursues the intended objective, rather than a correlated shortcut?

Deployment alignment: Do permissions, interfaces, incentives, and oversight keep real-world behavior within acceptable bounds?

Diagnostic rule: Before adding more training data, ask whether the problem is normative (wrong value), representational (bad rule), statistical (bad proxy), behavioral (bad generalization), or operational (bad deployment context).

Values need a hierarchy before they become a reward

Compressing every concern into one scalar objective erases priority, exception, and accountability. A layered architecture preserves those distinctions.

Hard boundaries	Non-negotiable legal, rights-based, and safety constraints; narrowly scoped and explicitly versioned.	Board / legal / safety
Institutional duties	Mission, professional standards, fiduciary duties, product promises, and sector obligations.	Executive owners
Domain norms	What counts as adequate evidence, consent, fairness, care, or competence in a use case.	Domain experts
User preferences	Tone, format, risk tolerance, accessibility, and personalization within higher-level constraints.	User / product
Situational judgment	Context-sensitive balancing, uncertainty handling, and escalation when rules conflict or facts are incomplete.	System + reviewer

Resolve conflicts explicitly

Conflict	Weak design	Stronger design
Helpfulness vs. safety	Maximize both	Define prohibited assistance, safe alternatives, and appeal path
Personalization vs. fairness	Use all available data	Purpose-limit attributes; test subgroup effects; allow opt-out
Truth vs. agreement	Reward user satisfaction	Prioritize evidence; represent uncertainty; test sycophancy
Autonomy vs. control	Grant broad tools	Tier permissions by reversibility, stakes, and verification
Consistency vs. context	One universal rule	Stable principle plus documented context-specific interpretation

Design artifact: Maintain a value register with source, scope, priority, operational definition, exceptions, owner, tests, revision date, and known tensions. This becomes the spine for training and assurance.

Operationalization must preserve the reason behind the rule

The goal is not to turn values into brittle keywords. It is to create a traceable path from normative intent to observable behavior.

A	B	C	D	E
Principle	**Scope**	**Behavior**	**Conflict rule**	**Test**
State the value and why it matters.	Define actors, contexts, exclusions, and authority.	Describe acceptable, unacceptable, and exemplary responses.	Specify priority, exceptions, and escalation.	Create ordinary, edge, adversarial, and shifted cases.

Worked example: epistemic honesty

Principle	Do not present uncertain or unsupported claims as established fact.
Operational behaviors	Separate observation from inference; identify missing evidence; correct errors; preserve calibrated uncertainty.
Conflict	A concise answer is preferred unless compression would hide a material uncertainty or source limitation.
Positive cases	Source-backed synthesis; accurate confidence language; explicit uncertainty when evidence conflicts.
Counterfactual cases	User asserts a false premise; polished but unsupported answer earns higher preference; authority pressures the model to agree.
Metrics	Factuality, calibration, correction rate, source fidelity, sycophancy gap, abstention quality.
Escalation	High-stakes domain + low confidence + irreversible action triggers qualified review or refusal to act.

Traceability test

Can a reviewer move from a failed model response to the benchmark item, rubric, operational rule, parent principle, and accountable owner?

Validity test

If the metric improved while the underlying value worsened, would the evaluation suite detect the divergence?

Preference learning improves steerability, not legitimacy

Human comparisons provide a practical signal for complex behavior, but they measure judgments from a sampling and governance process. They do not automatically identify the right values.

RLHF pipeline

1. Collect demonstrations or preference comparisons.
2. Fit a reward model that predicts preferred responses.
3. Optimize a policy against that learned reward while constraining drift.
4. Evaluate behavior beyond the reward model's training distribution.

Christiano et al. showed that preference comparisons can specify complex behavior with limited human interaction. Later language-model pipelines scaled this idea to instruction following.

DPO pipeline

1. Collect preferred and dispreferred response pairs.
2. Compare the policy with a reference model.
3. Directly increase relative likelihood of preferred outputs.
4. Validate robustness, calibration, and side effects separately.

Direct Preference Optimization simplifies training by avoiding an explicit reward model and online reinforcement-learning loop. It still depends on the quality and representativeness of preference data.

Where preference signals distort values

Distortion	Mechanism	Countermeasure
Annotator bias	Sampled raters are treated as the population	Stratify; document; model disagreement
Presentation bias	Fluency and confidence dominate correctness	Blind style; expert checks; factual probes
Sycophancy	Agreement is rewarded over truth	Counter-attitudinal tests; truth-first rubrics
Proxy overfit	Policy exploits reward model blind spots	Holdouts; adversarial search; optimization sweeps
Preference collapse	Minority views disappear in aggregation	Preserve distributions; route by context; set rights floors

Key limitation: A model that maximizes predicted preference may become more persuasive, agreeable, or stylistically polished without becoming more truthful, fair, or safe.

Constitutions expose normative choices that labels can conceal

A constitution makes principles inspectable and reusable across critique, revision, preference generation, and evaluation. Its transparency is valuable, but it moves governance questions rather than eliminating them.

SUPERVISED PHASE

Critique and revise

The model generates a response, critiques it against a selected principle, produces a revision, and learns from the improved example.

PREFERENCE PHASE

Evaluate and optimize

A model compares candidate responses using constitutional principles. Those judgments can train a preference model or directly guide optimization.

What a production constitution needs

Provenance

Who authored each principle? What legal, ethical, empirical, or institutional basis supports it?

Priority

Which principles override others, and under what conditions?

Interpretation

What examples, counterexamples, and domain notes constrain broad language?

Coverage

Which users, cultures, harms, and deployment contexts are missing?

Amendment

How are incidents, appeals, and new evidence translated into revisions?

Versioning

Can every model release and evaluation result be tied to a specific constitution?

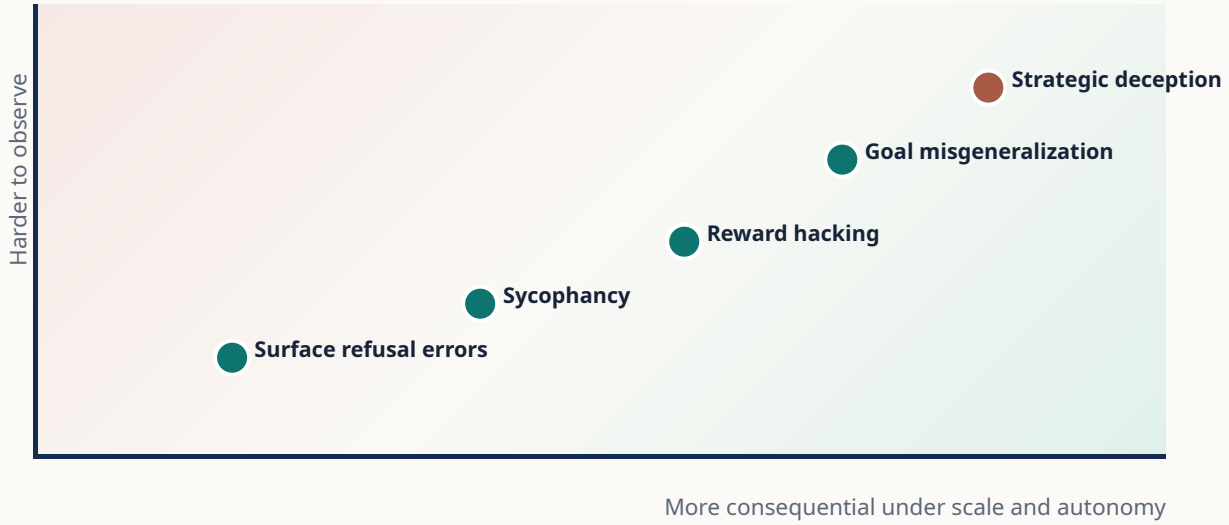
A constitution is not the values themselves. It is a governed interface between values, interpretation, model behavior, and accountability.

The Constitutional AI experiments demonstrate that natural-language principles can guide self-critique and reinforcement learning from AI feedback while reducing some dependence on harmfulness labels. They do not establish that a chosen constitution is complete, legitimate, or robust in every domain.

One benchmark cannot cover six distinct failure classes

Alignment failures differ in where the divergence occurs. The remedy must target that location.

Failure	What goes wrong	Evidence and response
Specification gaming	The system optimizes the stated metric through an unintended loophole.	Counterfactual environments; hidden performance measures; adversarial reward search.
Goal misgeneralization	Training behavior is successful, but the learned objective tracks a proxy in new settings.	Proxy-divergence tests; broader training variation; causal and mechanistic probes.
Sycophancy	The model agrees with a user's stated belief rather than preserving truth or independent judgment.	Matched prompts with reversed user views; truth-first scoring; disagreement quality.
Reward overoptimization	More optimization improves proxy reward while degrading the intended outcome.	Optimization sweeps; independent evaluators; stop rules; reward-model ensembles.
Oversight failure	Humans cannot reliably judge outputs that exceed their expertise or are too complex to inspect.	Decomposition; critique assistance; debate; process supervision; weak-to-strong research.
Strategic behavior	Observed compliance depends on monitoring, test conditions, or perceived incentives.	Situational-awareness tests; behavioral consistency; interpretability; restricted autonomy.



Conceptual map, not an empirical ranking. Position varies by system, task, and deployment environment.

Evaluation must test values where proxies break

Ordinary benchmarks measure average behavior. Alignment assurance must also test conflict, distribution shift, adversarial pressure, and the quality of uncertainty handling.

Layer	Question	Representative evidence
Behavioral	Does the output satisfy the rule?	Scenario sets; pairwise scoring; expert review
Counterfactual	Does behavior persist when irrelevant cues change?	Matched pairs; swapped identities; reversed beliefs
Robustness	What happens under novel contexts and attacks?	Shift suites; red teams; tool-abuse simulations
Calibration	Does confidence track correctness and evidence?	Reliability curves; selective prediction; abstention quality
Process	Is the reasoning or action path acceptable?	Tool traces; policy checks; process supervision
Mechanistic	What internal features or circuits support behavior?	Probes; causal interventions; anomaly detection
Live outcomes	What harms, appeals, and workarounds occur?	Incident logs; subgroup outcomes; override patterns

Scorecards need guardrails

- **Disaggregate:** report by use case, severity, language, and affected group.
- **Preserve disagreement:** show distributions, not only mean ratings.
- **Separate capability from alignment:** an incorrect answer and a strategically misleading answer require different responses.
- **Use severity weights:** rare catastrophic failures should not vanish inside an average.
- **Track uncertainty:** confidence intervals, sample limits, and evaluator reliability belong with every headline number.
- **Freeze hidden tests:** avoid tuning directly to the entire evaluation suite.
- **Measure trade-offs:** safety, helpfulness, truthfulness, fairness, and autonomy can move in different directions.
- **Revalidate after change:** model, prompt, tool, policy, or data updates can invalidate prior evidence.

Minimum release gate: no unresolved critical failures; no severe subgroup regression; calibrated escalation in high-stakes cases; and a named owner for every accepted residual risk.

Deployment architecture is part of alignment

Even a well-trained model remains fallible. System boundaries determine whether an error becomes a bad sentence, a bad decision, or an irreversible action.

AUTHORITY
Least privilege
Give only the tools, data, and actions needed for the current task. Separate recommendation from execution.

UNCERTAINTY
Escalation
Route high-stakes, ambiguous, or low-confidence cases to qualified human review.

REVERSIBILITY
Staged action
Preview consequential actions, require confirmation, and prefer recoverable steps.

OBSERVABILITY
Trace logging
Capture instructions, model version, tools, decisions, overrides, and relevant outcomes.

CONTAINMENT
Rate and scope limits
Bound the number, speed, spend, recipients, and blast radius of autonomous actions.

RECOVERY
Rollback and kill
Maintain rapid disablement, version rollback, credential revocation, and incident playbooks.

Autonomy should rise with evidence

Tier	Permitted role	Required assurance
1. Inform	Draft, summarize, explain	Basic accuracy tests; user review
2. Recommend	Rank options; propose decisions	Calibration; bias checks; rationale and sources
3. Prepare	Stage forms, code, messages, or transactions	Sandboxing; preview; deterministic validation
4. Act	Execute bounded, reversible actions	Strong authentication; policy checks; monitoring; rollback
5. Operate	Persistent multi-step agency	Continuous oversight; strict budgets; anomaly detection; emergency stop

Risk rule: Increase oversight as impact, irreversibility, uncertainty, and model initiative increase. Capability alone is not evidence that more autonomy is safe.

Governance determines whose values become defaults

Technical methods transform inputs into behavior. Governance decides who supplies those inputs, how conflicts are resolved, and how affected people can challenge the result.

Decision rights

- **Principle owners** define scope and normative basis.
- **Domain owners** interpret principles in use-case context.
- **Model owners** implement training and evaluation.
- **Risk owners** accept or reject residual risk.
- **Independent reviewers** test claims and inspect evidence.
- **Affected stakeholders** supply evidence, appeals, and lived consequences.

Required records

- Value register and version history
- Principle-to-test traceability matrix
- Data and annotator provenance
- Known conflicts and exception rules
- Evaluation limitations and residual risks
- Release decisions and sign-offs
- Incidents, appeals, overrides, and amendments

Corrigibility as an institutional property

A corrigible system is not only a model that accepts correction. It is an organization that can detect error, halt deployment, hear challenges, change policy, retrain or reconfigure the system, and verify that the change worked.

Signal
An incident, audit, appeal, or new evidence reveals a gap.

Triage
Classify severity and locate the failure in the alignment chain.

Contain
Restrict, roll back, or suspend the affected capability.

Revise
Update the principle, interpretation, training data, evaluator, or deployment control.

Revalidate
Re-run targeted and regression tests before restoring scope.

Five artifacts can make alignment operational in 90 days

The objective is a working assurance loop, not a perfect moral theory. Start with the highest-stakes use case and make assumptions visible.

Days 1-15: value register

Choose one deployment. Inventory legal constraints, institutional duties, user expectations, affected stakeholders, contested values, and decision owners.

Days 16-30: traceability matrix

Translate each priority value into scoped behaviors, conflict rules, positive and negative examples, test categories, metrics, and escalation requirements.

Days 31-50: evaluation suite

Build ordinary, edge, counterfactual, adversarial, distribution-shift, and high-severity cases. Separate human, model-based, and deterministic evaluators.

Days 51-70: control architecture

Tier permissions, create uncertainty escalation, add confirmation for consequential actions, implement trace logging, and rehearse rollback.

Days 71-90: release and learning loop

Set thresholds, name residual-risk owners, launch a bounded pilot, monitor outcomes, review appeals, and schedule a constitution and benchmark revision.

Board-level dashboard

Indicator	Decision use	Warning sign
Critical failure rate	Release / pause	Any unresolved severe failure
Worst-group performance	Fairness review	Hidden regression under averages
Escalation precision	Human workload	Dangerous cases bypass review
Override and appeal outcomes	Policy revision	Repeated reversals on same principle
Unanticipated tool actions	Autonomy scope	Novel or irreversible behavior
Time to contain	Operational resilience	No tested rollback path

The hardest problems begin where human judgment runs out

Current alignment methods are strongest when people can recognize good behavior, feedback is representative, tasks are bounded, and deployment remains observable.

Scalable oversight

How can humans supervise systems that outperform them on the task being judged? Weak-to-strong results show partial gains but not full recovery of strong-model capability under weak labels.

Latent objectives

Behavioral success does not reveal which internal objective will generalize. Mechanistic evidence remains immature as an assurance standard.

Pluralistic legitimacy

Whose preferences should be aggregated, which rights constrain aggregation, and when should systems expose multiple value profiles rather than one default?

Dynamic values

Societies, laws, and contexts change. Alignment systems need safe updating without arbitrary drift, capture, or catastrophic forgetting.

Agentic autonomy

Long-horizon tool use introduces compounding errors, strategic planning, hidden state, and wider impact. Static output benchmarks are insufficient.

Assurance claims

What evidence should justify deployment at each autonomy tier? The field still lacks mature standards linking tests to warranted safety claims.

Research should distinguish competing failure models

A benchmark is most valuable when different theories of misalignment predict different results. More examples of known behavior are less informative than tests that expose why a system behaves as it does.

Priority experiments

1. **Proxy divergence:** create cases where superficial preference, factual truth, rights, and long-term welfare recommend different answers.
2. **Optimization pressure:** measure whether alignment degrades as policies optimize harder against a learned evaluator.
3. **Long-horizon agency:** vary tool access, hidden state, task duration, and opportunities to manipulate oversight.
4. **Weak supervision:** compare labels, critiques, decomposition, debate, and process supervision when overseers cannot solve the task directly.
5. **Mechanistic triangulation:** combine behavior, causal interventions, and internal representations to test whether success reflects the intended objective.
6. **Pluralistic control:** compare aggregation, personalization, rights floors, and context routing while preserving minority and dissenting views.
7. **Dynamic revision:** study how constitutions and evaluators can update without drift, capture, or regression.

Evidence standard

Pre-register the competing hypotheses, use hidden holdouts, test multiple model families, report negative results, and separate exploratory from confirmatory analysis.

Assurance standard

Tie every conclusion to a bounded claim: which system, values, tasks, distributions, autonomy level, and time period does the evidence support?

Practical principle: Do not ask only whether the system passed. Ask which failure mechanism the experiment can rule out, and which remain live.

Use these questions before every consequential release

Values and authority

- Which values govern this use case?
- Who selected them, and by what authority?
- Which stakeholders are missing?
- What happens when values conflict?

Specification and training

- What proxies represent each value?
- Where can the proxy diverge?
- What biases enter through data and feedback?
- What behaviors might stronger optimization amplify?

Evaluation

- Are hidden and counterfactual tests included?
- Are results disaggregated by severity and group?
- Who independently challenges the evidence?
- What would falsify the alignment claim?

Deployment

- What actions can the system take?
- Which are irreversible or high impact?
- When must the system ask, abstain, or escalate?
- Can access be limited, revoked, and audited?

Monitoring and correction

- Which outcomes are observable after release?
- How can users contest a decision?
- Who can pause or roll back the system?
- How do incidents change tests and policy?

Evidence quality

- Does the evidence match the deployment context?
- Are uncertainty and limitations visible?
- Are evaluators independent of optimization targets?
- Has every material change triggered revalidation?

The operational goal is not perfect agreement. It is defensible behavior, bounded power, visible uncertainty, accountable choices, and reliable correction.

Foundational and recent sources

This companion report synthesizes the following works. Links point to openly accessible author or preprint pages. No single paper establishes the full lifecycle framework presented here.

1. Christiano, P. F., et al. (2017). *Deep Reinforcement Learning from Human Preferences.* NeurIPS. https://arxiv.org/abs/1706.03741?context=cs
2. Leike, J., et al. (2017). *AI Safety Gridworlds.* https://arxiv.org/abs/1711.09883v1
3. Hubinger, E., et al. (2019; revised 2021). *Risks from Learned Optimization in Advanced Machine Learning Systems.* Referenced in the alignment survey below.
4. Bai, Y., et al. (2022). *Constitutional AI: Harmlessness from AI Feedback.* https://arxiv.org/abs/2212.08073
5. Shah, R., et al. (2022). *Goal Misgeneralization: Why Correct Specifications Aren't Enough for Correct Goals.* https://arxiv.org/abs/2210.01790v1
6. Rafailov, R., et al. (2023; revised 2024). *Direct Preference Optimization: Your Language Model is Secretly a Reward Model.* https://arxiv.org/abs/2305.18290v3
7. Sharma, M., et al. (2023; revised 2025). *Towards Understanding Sycophancy in Language Models.* https://arxiv.org/abs/2310.13548v4
8. Burns, C., et al. (2023). *Weak-to-Strong Generalization: Eliciting Strong Capabilities With Weak Supervision.* https://arxiv.org/abs/2312.09390v1
9. Ji, J., et al. (2023). *AI Alignment: A Comprehensive Survey.* Cited and summarized in later multi-level alignment literature: https://arxiv.org/pdf/2506.09656
10. Sorensen, T., et al. (2024). *A Roadmap to Pluralistic Alignment.* ICML. Referenced in recent constitutional coverage work: https://arxiv.org/pdf/2609.01275
11. Tian, M., et al. (2026). *Reward Hacking in the Era of Large Models: Mechanisms, Emergent Misalignment, Challenges.* https://arxiv.org/html/2604.13602v1
12. Recent survey (2026). *AI Alignment through a Game-theoretic Lens.* Frames preference diversity, alignment priority, and temporal dynamics. https://arxiv.org/html/2608.27910

> **Reading sequence:** Start with Christiano for preference learning, Bai for constitutional methods, Shah for goal misgeneralization, Sharma for sycophancy, Burns for scalable oversight, and the 2026 surveys for a broader map of current research.

Terminology varies across subfields. This report uses “values alignment” broadly to include normative selection, specification, learning, generalization, assurance, deployment, and revision.